NATIONAL UNIVERSITY BANGLADESH



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Syllabus

Bachelor of Science (Honours) Second Year

Department of Soil Science

Effective from the Academic Year: 2024–2025 and onward

1. Vision:

We envision the National University as a truly national institution where our commitment to **"Progress for Everyone"** is reflected across all areas of our operations and engagement with stakeholders. We will harness our core strengths in education, research, innovation, and community engagement to address pressing national and global challenges and contribute to sustainable development. The National University aspires to consolidate its position as one of Bangladesh's leading academic institutions and establish a strong presence among the world's renowned universities. Our strategic priorities will focus on expanding academic and research expertise, building strong national and international partnerships, fostering innovation, and enhancing our global academic and research influence. The quality and impact of our services will ultimately be measured by the achievements, opportunities, and improved quality of life experienced by the individuals, institutions, and communities we serve.

2. Mission

To advance **"Progress for Everyone"** through transformative education, innovative and impactful research, and meaningful collaboration with individuals, communities, and national and international partners.

Name of the Program : B.Sc. (Honours) in Soil Science

3. Programme Educational Objectives (PEOs):

PEO 1: Core Scientific Competency

To develop a strong foundation in the fundamental principles of soil science, including soil genesis, classification, soil physics, soil chemistry, soil fertility, soil microbiology, and environmental interactions, to understand soil as a natural resource critical for sustainable agriculture and ecosystem.

PEO 2: Application in Agriculture and Environment

To apply soil science knowledge in solving real-world problems related to sustainable agriculture, land use, environmental management, and natural resource conservation in Bangladesh and beyond.

PEO 3: Research and Higher Studies

To pursue advanced education, research, and innovation in soil science and allied disciplines such as agronomy, crops of Bangladesh, environmental science, geology, and climate change adaptation.

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PEO 4: Ethical and Social Contribution

To demonstrate ethical responsibility and commitment to serving society by promoting sustainable use of soil resources, environmental awareness, and contributing to national development goals such as food security and climate resilience.

PEO 5: Lifelong Learning and Professional Development

To develop the capacity for lifelong learning, continuous professional development, innovation, and adaptation to emerging technologies and interdisciplinary approaches in soil science and related fields, enabling graduates to contribute effectively to national and global sustainable development.

4. Programme Learning Outcomes (PLOs)

All the time of graduation B.Sc.(Honours) in Soil Science students will have achieved the ability to :

Programme Learning Outcomes (PLOs)

PLO 1: Knowledge and Understanding of Soil Science

Demonstrate comprehensive knowledge and understanding of soil formation, classification, chemistry, fertility, biology, physics, and soil–environment interactions.

PLO 2: Teamwork, Leadership and Professional Skills

Work effectively as an individual, team member or leader in multidisciplinary environments and engage in lifelong learning and continuous professional development to adapt to emerging technologies, scientific advancements, and changing societal needs

PLO 3: Environmental Responsibility, Ethics and Professionalism

Demonstrate environmental stewardship, sustainable land-use practices, academic integrity, scientific ethics, professional responsibility, and awareness of the social responsibilities of soil scientists.

PLO 4: Research, Inquiry and Communication

Design, conduct, analyze, and interpret scientific research using appropriate methodologies and ethical practices, and communicate scientific information and research findings effectively through oral, written, graphical, and digital formats.

PLO 5: Laboratory, Field and Technical Skills

Demonstrate proficiency in soil sampling, laboratory analysis, field techniques, soil testing, instrumentation, data analysis, and relevant digital technologies and software.

PLO 6: Application, Problem Solving and Critical Thinking

Apply scientific principles, critical thinking, and evidence-based approaches to solve complex problems related to soil, agriculture, environmental protection, land management, and natural resource sustainability

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5. Mapping PEO with PLO:

PEO / PLO	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6
PEO 1	✓			✓	✓	✓
PEO 2	✓		✓		✓	✓
PEO 3	✓	✓		✓	✓	✓
PEO 4		✓	✓	✓		✓
PEO 5	✓	✓	✓	✓	✓	✓

6. Courses, Marks and Credit Distribution

SECOND YEAR

Course Code	Course Title	Marks	Credits
8223351	Soil Physics-1	100	4
8223353	General Microbiology	100	4
8223355	Soil Mineralogy and colloids	100	4
8223366	Soil Science Practical	100	4
8222857	Chemistry-II	100	4
8222861	Environmental Chemistry	50	2
8223057	Botany – II	100	4
8223058	Botany Practical – II	50	2
8221171	Functional English	100	4
8226601	Applied Information and Communication Technology (Applied ICT)	75	3
8226602	Practical: Applied ICT	25	1
8223358	Viva voce (Soil Science)	50	2
Total		**950**	**38**

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Detailed Syllabus

Course Introduction

Course Code	8223351	Marks : 100	Credits : 4	Class Hours : 56
Course Title	Soil Physics – I			

Rationale of the Course

This course provides a comprehensive foundation in soil physical properties and processes that govern soil behavior, plant growth, water movement, aeration, and agricultural productivity. It covers key topics including particle-size distribution, soil texture and structure, soil consistence, soil air, thermal regime, soil water and related properties**.** Understanding these properties is essential for assessing soil quality, managing soil water and aeration, optimizing crop production, and promoting sustainable soil and environmental management**.**

Course Objectives

The objective of this course is to provide students with a fundamental understanding of the physical properties and processes of soil**,** including particle-size distribution, texture, structure, and consistence. The course emphasizes soil air, water, thermal regime properties, with particular attention to their measurement, interpretation, and practical significance. It aims to develop students' ability to assess soil physical conditions and understand their influence on plant growth, water movement, aeration, and agricultural productivity.

Ultimately, the course equips students with the knowledge and skills to apply soil physical principles **in** effective, efficient, and sustainable soil management and mapping with SDG. This course is relevant to achieve SDG 4.

Course Learning Outcomes (CLOs) and Mapping with PLOs

Upon completion of this course, the students will be able to achieve the following learning outcomes:

CLO	Course Learning Outcomes	PLO Covered
CLO 1	Explain basic physical properties of soil related to its solid phase, including particle-size distribution, soil textural classes, and density.	PLO 1
CLO 2	Describe soil structural classification, genesis, Atterberg's constants (consistence), and evaluate their agricultural significance.	PLO 1, PLO 6
CLO 3	Analyze the composition of soil air, gas transport, gaseous exchange mechanisms, and apply techniques to measure soil aeration.	PLO 1, PLO 5
CLO 4	Evaluate the soil thermal regime, atmospheric energy balance, and determine thermal properties of soil (heat capacity, thermal conductivity).	PLO 1, PLO 4
CLO 5	Determine soil water classifications, measurement methods, and application approaches to address environmental and land management challenges.	PLO 3, PLO 5, PLO 6
CLO 6	Work effectively in teams to conduct field and laboratory analyses on soil physical properties and communicate scientific findings in an integrated way.	PLO 2, PLO 4

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Course Contents:

S.L.	Topic	Class Hours	CLO Covered
1	**Physical Nature of Soil Particles:** Primary knowledge on soil physical properties; particle-size distribution, soil textural classes; Soil bulk density and particle density.	10	CLO 1
2	**Soil Structure & Consistence:** Soil Structure - classification, genesis, evaluation and agriculture significance; Soil Consistence – Atterberg's constants and their practical significance.	10	CLO 2
3	**The Gaseous Phase:** Composition of soil air, gas transport through soil, gaseous exchange - the renewal of soil air; measurement of soil aeration.	10	CLO 3
4	**The Soil Thermal Regime:** Atmospheric energy balance, solar radiation, factors affecting soil temperature, soil surface energy balance.	10	CLO 4
5	**Thermal Properties of Soil:** Temperature, heat capacity and thermal conductivity, measurement of thermal conductivity.	8	CLO 4
6	**Soil Water potentials:** Soil Water – Classifications and determination and approaches and applications.	8	CLO 5

Mapping CLOs with Teaching-Learning and Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO 1	Visual and Auditory	Class attendance, In-course Exam, Assignment, Final theory and practical Exam, Field level assessment.
CLO 2	Visual, Auditory and Kinesthetic	
CLO 3-6	Visual and Auditory	

Visual : Learning by seeing (pictures, diagrams, reading and charts).
Auditory : Learning by hearing (discussions, lectures and listening)
Kinesthetic : Learning by doing (physical movement, touch and hands-on practice)

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

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Learning Materials:

Baver, L. D., Gardner, W. H., & Gardner, W. R. (1972). *Soil physics* (4th ed.). John Wiley & Sons.

Hillel, D. (1998). *Environmental soil physics: Fundamentals, applications, and environmental considerations.* Academic Press.

Kohnke, H. (1979). *Soil physics*. Tata McGraw-Hill Publishing Company.

Scott, H. D. (2000). *Soil physics: Agricultural and environmental applications***.** Iowa State

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Course Code	8223353	Marks : 100	Credits : 4	Class Hours : 56
Course Title	**General Microbiology**			

Rationale of the Study

This course provides basic knowledge of microorganisms, soil microbial processes, microscopic study, aseptic techniques, microbial control, metabolism, and enzymes. It helps students understand microbial activities, nutrient cycling, soil fertility, and sustainable soil management**.**

Course Objectives:

The objective of this course is to provide students with a fundamental understanding of microbial diversity, classification, nutritional requirements, and the ecological roles of microorganisms in the soil environment**.** The course also introduces students to the basic principles and techniques of microscopic assessment, isolation and maintenance of pure cultures, aseptic practices, and microbial control**.** In addition, it covers microbial metabolism of carbohydrates, protein synthesis, enzyme functions, and oxidative and fermentative pathways, with particular emphasis on the mode of enzyme action, Michaelis–Menten constant (Km), and factors influencing enzyme activity. Mapping with SDG: This course is relevant to achieve SDG 4.

Course Learning Outcomes (CLOs) and Mapping with PLOs

Upon completion of this course, the students will be able to achieve the following outcomes:

CLO	Course Learning Outcomes	PLOs Covered
CLO 1	Explain the classification, diversity, characteristics, and nutritional requirements of microorganisms in relation to the living world.	PLO1
CLO 2	Describe the role of soil as a habitat, energy source, and nutrient reservoir for microorganisms and explain the importance of soil organic matter and biomass.	PLO1, PLO3
CLO 3	Explain the principles, applications, and care of microscopes and demonstrate basic skills in pure culture isolation and aseptic techniques.	PLO1, PLO5
CLO 4	Apply appropriate physical and chemical methods of microbial control and explain the use and action of antimicrobial agents.	PLO3, PLO5, PLO6
CLO 5	Explain microbial carbohydrate metabolism, oxidative and fermentation pathways, protein synthesis, and enzyme functions, including Km and factors affecting enzyme activity.	PLO1, PLO6
CLO 6	Conduct microbiological activities safely and ethically, work effectively in teams, and communicate experimental observations and scientific findings using appropriate formats.	PLO2, PLO4, PLO3

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Course Contents:

Sl.	Topics	Class Hours	CLOs Covered
1	Microorganisms & Classification: Microorganisms in the living world; two-kingdom classification; Haeckel's kingdom; Protista; eukaryotic and prokaryotic protists; Whittaker's five-kingdom concept; nutritional requirements.	8	CLO 1
2	The Soil Environment and hub of bioenergy: Soil as a source of energy and nutrients for biota; soil organic matter and living biomass as sources of nutrients.	6	CLO 2
3	Microscope and its components: Principles, functions, applications and care of different types of microscopes.	10	CLO 3
4	Isolation, Aseptic Techniques & Microbial Control: Isolation of pure cultures from different sources; aseptic techniques; physical and chemical methods of microbial control; antimicrobial agents.	12	CLO 4
5	Microbial Metabolism: Microbial metabolism of carbohydrates; oxidative and fermentation pathways; enzymes involved in metabolic processes.	10	CLO 5
6	Protein Synthesis & Enzymes: Protein synthesis; types and functions of enzymes; mode of enzyme action; Km; factors affecting enzyme activity.	10	CLO 5

Mapping CLOs with Teaching-Learning and Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO 1-2	Visual and Auditory	Class attendance, In-course Exam, Assignment, Final theory Exam, lab investigations, Group level assessment.
CLO 3-4	Visual, Auditory and Kinesthetic	
CLO 5	Visual and Auditory	
CLO 6	Visual, Auditory and Kinesthetic	

Visual : Learning by seeing (pictures, diagrams, reading and charts).
Auditory: Learning by hearing (discussions, lectures and listening)
Kinesthetic: Learning by doing (physical movement, touch and hands-on practice)

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Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Atlas, R. M. (2010). *Principles of microbiology*. ASM Press.

Dubey, R. C. (2011). *A textbook of microbiology*. S. Chand Publishing.

Pelczar, M. J., Chan, E. C. S., & Krieg, N. R. (1986). *Microbiology* (5th ed.). McGraw-Hill

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Course Code	8223355	Marks : 100	Credits : 4	Class Hours : 56
Course Title	**Soil Mineralogy and Soil Colloids**			

Rational of the Course:
This course provides a fundamental understanding of soil minerals, clay minerals, and soil colloids**,** focusing on their structure, formation, properties, charge behavior, and surface reactions. It provides essential knowledge of cation exchange, adsorption–desorption, and colloidal interactions for understanding soil fertility, nutrient dynamics, soil quality, and sustainable soil management.

Course Objectives:

The objective of this course is to provide students with a comprehensive understanding of soil minerals, clay minerals, and their physicochemical properties**,** including their formation, structure, stability, identification, and distribution. The course emphasizes clay charge, cation exchange capacity, soil oxides and hydroxides, and amorphous materials**,** along with the nature, classification, properties, and stability of soil colloids and their distinction from suspensions and solutions. The course also covers adsorption and desorption processes, adsorption capacity, and adsorption isotherms and equations**,** with emphasis on their applications in understanding soil colloidal processes and nutrient and chemical retention.
Mapping with SDG: This course is relevant to achieve SDG 4.

Course Learning Outcomes (CLOs) and Mapping with PLOs:

Upon completion of this course, the students will be able to achieve the following outcomes:

CLO	Course Learning Outcomes	PLOs Covered
CLO1	**C**lassification, characteristics, crystal chemistry, stability, and sequence of occurrence of primary and secondary silicate and non-silicate minerals in different soil size fractions.	PLO1
CLO2	**T**he formation, crystal structure, physicochemical properties, and identification methods of major clay minerals, oxides, hydroxides, and amorphous soil constituents.	PLO1, PLO2
CLO3	**T**he origin and development of surface charges in clay minerals, cation exchange capacity (CEC), and the properties of oxide and hydroxide minerals, with reference to their distribution in major soils of Bangladesh.	PLO1, PLO6
CLO4	**T**he fundamental properties of colloidal systems from suspensions and true solutions, including kinetic, optical, hydration, swelling, flocculation, and iso-electric properties.	PLO1, PLO2
CLO5	**The** electro-kinetic and membrane properties of soil colloids using Perrin's law, electrical double/triple layer concepts and Donnan membrane equilibrium in relevant soil systems.	PLO1, PLO6
CLO6	**S**tandard laboratory and field techniques using appropriate analytical tools for mineral identification, clay characterization, colloidal stability assessment, and soil testing, while maintaining accuracy and laboratory safety.	PLO3, PLO 5
CLO7	**Applying** adsorption–desorption principles and mathematical adsorption equations to determine adsorption capacity and interpret surface behavior of different soil colloidal components.	PLO4, PLO6

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Course Contents:

SL.	Topic/Key contents	Class Hours	CLOs Covered
1	Introduction to Soil Mineralogy and Primary Minerals: Concept and scope of soil mineralogy; importance of minerals in soils; classification of soil minerals; primary and secondary minerals; silicate and non-silicate minerals; occurrence of minerals in different soil size fractions; classification, crystal structure and chemistry of primary silicate minerals; quartz, feldspars, micas, pyroxenes, amphiboles and olivine; stability and sequence of mineral occurrence during weathering	6	CLO1
2	Secondary and Non-Silicate Minerals: Formation and characteristics of secondary minerals; carbonates, phosphates, sulfates and other non-silicate minerals; mineral stability and weathering sequence; distribution of minerals in different soil fractions; relationship between mineral composition and soil properties	6	CLO1, CLO2
3	Clay Minerals: Formation, Classification and Crystal Structure: Origin and formation of clay minerals; 1:1, 2:1 and 2:1:1 layer structures; kaolinite, illite, smectite, vermiculite and chlorite; crystal structure, layer arrangement, interlayer materials and major physicochemical properties	6	CLO2
4	Clay Mineral Chemistry, Charge and Identification: Isomorphous substitution; permanent and variable charge; layer charge; hydration and swelling; shrink–swell behavior; methods of clay-mineral identification including XRD, DTA and selective dissolution; interpretation of mineral identification data	6	CLO2, CLO3, CLO7
5	Soil Colloids and Their Fundamental Properties: Definition and classification of soil colloids; colloids compared with true solutions and suspensions; kinetic and optical properties; Brownian movement; Tyndall effect; hydration; swelling; flocculation; dispersion; isoelectric properties and colloidal stability	6	CLO4
6	Surface Charge, Ion Exchange and CEC: Origin and development of permanent and variable charges; pH-dependent charge; point of zero charge; cation exchange capacity (CEC); anion exchange capacity; ion adsorption and exchange reactions; role of clay minerals and oxides in charge development and ion exchange	6	CLO3, CLO4, CLO7
7	Electrical Double Layer and Electro-Kinetic Properties: Electrical double layer concepts; Helmholtz, Gouy–Chapman and Stern models; diffuse double layer; zeta potential; electro-osmosis; electrophoresis; electro-kinetic behavior of soil colloids;	10	CLO5
8	Adsorption–Desorption and Surface Reactions: Adsorption and desorption mechanisms; physical and chemical adsorption; adsorption sites and surface reactions; adsorption of ions by clay and oxide colloids; adsorption isotherms; Freundlich and Langmuir equations; determination and interpretation of adsorption capacity; application to soil and environmental management	10	CLO6, CLO7, CLO8

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Mapping CLOs with Teaching-Learning and Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO 1-8	Visual, Auditory and Kinesthetic	Class attendance, In-course Exam, Assignment, Final theory Exam, Group level assessment.

Visual : Learning by seeing (pictures, diagrams, reading and charts).
Auditory : Learning by hearing (discussions, lectures and listening)
Kinesthetic : Learning by doing (physical movement, touch and hands-on practice)

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Bahl et al. 2009. Essentials of Physical Chemistry. S. Chand & Company Ltd., India.

Grim, R. E. Clay Mineralogy. McGraw-Hill Book Company, Inc. New York, USA.

Tan, K. H. 1993. Principles of Soil Chemistry. (2nd ed.) Marcel Dekker, Inc.

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Course Introduction:

Course Code	223356	Marks : 100	Credits : 4	Class Hours : 56
Course Title	**Soil Science Practical**			

Rationale of the Course:

The course provides essential hands-on laboratory and analytical skill**s** for understanding the physical, chemical, and biological properties of soil. Through practical determination of glucose, mineral identification, soil pH, free carbonates, cation exchange capacity, particle-size distribution, bacterial staining, and total and available nitrogen, students develop their ability to analyze soil properties, interpret laboratory results, and relate analytical findings to soil fertility and soil management**.**

Course Objectives:

The course aims to develop students' practical skills in soil and microbiological analysis, enabling them to determine and interpret important soil properties such as pH, free carbonates, cation exchange capacity, particle-size distribution, minerals, glucose, and total and available nitrogen, while applying appropriate laboratory techniques, calculations, safety procedures, and scientific reporting skills.

Mapping with SDG: This course is relevant to achieve SDG 4, 6, 13, 14, 15 and 17.

Course Learning Outcomes (CLOs) and Mapping with PLOs:

Upon completion of this course, the students will be able to achieve the following outcomes:

CLO	**Course Learning Outcome**	**PLOs Covered**
CLO1	**Identify** common soil minerals based on their observable physical characteristics and diagnostic properties.	PLO1, PLO5
CLO2	**Determine** glucose content using appropriate laboratory procedures and interpret the obtained results.	PLO4, PLO5, PLO6
CLO3	**Measure and interpret** soil pH and free carbonate content using standard laboratory methods.	PLO1, PLO5, PLO6
CLO4	**Determine** the cation exchange capacity (CEC) of soils and explain its significance in soil nutrient retention.	PLO1, PLO5, PLO6
CLO5	**Determine** soil particle-size distribution through practical size/mechanical analysis and interpret soil textural characteristics.	PLO1, PLO5, PLO6
CLO6	**Perform** bacterial staining techniques and recognize basic bacterial characteristics under a microscope.	PLO2, PLO5
CLO7	**Determine** total and available nitrogen in soil using appropriate analytical procedures and interpret their implications for soil fertility.	PLO1, PLO4, PLO5, PLO6
CLO8	**Apply** laboratory safety, accurate measurement, data recording, basic calculation, result interpretation, and scientific reporting skills in soil and microbiological practical work.	PLO2, PLO3, PLO4, PLO5

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Course Contents:

S.L	Topic / Key Contents	Class Hours	CLOs Covered
1	Course 223301**:** Determination of Soil pH: Principle, preparation, pH measurement, calibration, calculation and interpretation of soil reaction; Determination of Free Carbonates of Soil: Principle, reagents, experimental procedure, estimation, calculation and interpretation; Determination of CEC: Principle, sample preparation, laboratory procedure, calculation and significance of CEC in nutrient retention; Determination of Total Nitrogen: Principle, sample preparation, digestion/distillation or appropriate analytical procedure, calculation and interpretation; Determination of Available Nitrogen: Soil extraction, estimation, calculation, result interpretation and assessment of nitrogen availability; Particle Size Analysis of Soil: Particle-size analysis, dispersion, sedimentation/mechanical analysis, calculation, textural classification and interpretation.	**25**	CLO1 CLO3 CLO4 CLO5 CLO7 CLO8
2	Course 223303: Determination of Glucose: Principle, preparation of reagents, laboratory procedure, calculation, observation and interpretation of results; Bacterial Staining: Preparation of bacterial smear, fixation, staining procedure, microscopic observation and interpretation of bacterial characteristics.	12	CLO2, CLO6, CLO8
3	Course 223305: Identification of Soil Minerals: Physical characteristics, diagnostic properties and identification of common soil minerals.	10	CLO1, CLO2 CLO8

Mapping CLOs with Teaching-Learning and Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO 1-2	Visual, Kinesthetic	Practical test in lab and field as well, identification report preparation, practical identification, microscope observation, laboratory notebook, visual presentation and viva etc.
CLO 3-4	Visual, Auditory and Kinesthetic	
CLO 5	Visual, Kinesthetic-6	
CLO 7-8	Visual, Auditory and Kinesthetic	

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Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Black, C. A. (Ed.). (1965). Methods of soil analysis: Part 1. Physical and mineralogical properties. American Society of Agronomy.

Cappuccino, J. G., & Welsh, C. T. (2020). Microbiology: A laboratory manual (12th ed.). Pearson.

Jackson, M. L. (1973). Soil chemical analysis. Prentice-Hall of India.

Page, A. L., Miller, R. H., & Keeney, D. R. (Eds.). (1982). Methods of soil analysis: Part 2. Chemical and microbiological properties. American Society of Agronomy.

Sparks, D. L. (Ed.). (1996). Methods of soil analysis: Part 3. Chemical methods. Soil Science Society of America.

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Course Code: 8222859	Marks: 100	Credits: 4	Class Durations: 56
Course Title: Chemistry-II			

Course Description

This course provides a comprehensive foundation in the essential branches of chemical sciences, spanning biological, physical, inorganic, organic, industrial, and environmental chemistry. The curriculum begins with the fundamental chemical blocks of life, exploring structural and operational dynamics through bio-macromolecules, enzymes, co-enzymes, and endocrine hormones. In the realm of physical chemistry, it introduces the core principles of chemical energetics—focusing on the laws of thermodynamics, system boundaries, and enthalpy variations—alongside electrochemistry, exploring the mechanics, parameters, and electromotive forces (EMF) of galvanic and electrolytic cells. The inorganic module provides a periodic evaluation of \(s\)-, \(p\)-, and \(d\)-block elements and their industrial utility. Furthermore, the course expands into polymer chemistry, examining macromolecular structures, natural fibers, and biopolymers. Finally, it emphasizes environmental systems by evaluating pollution mechanics, persistent hazards like PCBs, biochemical toxicities of deadly heavy metals (mercury, chromium, arsenic, lead), and quantitative measurements of standard water quality indices such as pH, DO, BOD, and COD.

Course Objectives

Upon successful completion of this course, students will be able to:

- Elucidate the structural composition, synthesis pathways, functions, and biochemical operations of life-sustaining entities including carbohydrates, amino acids, proteins, enzymes, and hormones.
- Analyze thermodynamic properties, energy transformations, and work expansions using the laws of chemical energetics.
- Evaluate the electrical behaviors of electrochemistry systems by determining electrode potentials, EMF measurements, and cell mechanics for both reversible and irreversible operations.
- Explain the periodic trends, configurations, positions, and industrial application capacities of \(s\)-, \(p\)-, and \(d\)-block elements.
- Classify complex macromolecules, industrial polymers, elastomers, and biopolymers based on their synthesis, structure, and technological importance.
- Assess the societal and environmental footprint of pollutants by tracking persistent organic contaminants (PCBs), mitigating heavy metal toxicology, and executing standard parameters for soil and water quality monitoring.

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Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO	Course Learning Outcome	Bloom's Level
CLO 1	Explain the Chemistry of Life and Biomolecules	Understand
CLO 2	Analyze Chemical Energetics and Thermodynamic Systems	Evaluate
CLO 3	Evaluate Electrochemical Systems and Cells	Analyze
CLO 4	Categorize Main Group and Transition Elements	Understand
CLO 5	Characterize Polymers and Macromolecules	Understand
CLO 6	Assess Environmental Pollution and Toxicities	Analyze

Course Content

Week	Topic	Teaching Learning Strategy	CLOs
1	**1.The Chemistry of Life**: The stuff of life, Energy, Water, The chemical building block of life.	Lecture, Group Discussion, Assignment	1, 2, 3
3	**2.Chemical Energetics:** scopes and limitations of chemical thermodynamics, system, surrounding and boundary, heat and work, thermodynamic properties, state function, thermodynamic process, work of expansions reversible, irreversible and adiabatic expansions, concepts of internal energy and enthalpy, First law of thermodynamics: applications, entropy and its physical significance, free energy. various type of reaction enthalpy.	Lecture, Group Discussion, Assignment	3, 4, 5
3	**3.Electrochemistry:** Electrolytes, electrolysis: mechanism, factors and applications; electrodes, Electrolytic cell, Galvanic cells: Daniel Cell, simple voltaic cell, half-cell and half-cell reaction, cell notation, reversible & irreversible cell, emf and its measurement, standard emf.	Lecture, Group Discussion, Assignment	3, 4, 5
3	**4.s-block, p-block &d-block elements**: Overview of the main group elements, characteristics and periodic position of **s-, p- &d**-block elements, applications and	Lecture, Group Discussion, Assignment	2, 3, 4

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	important compounds, characteristics and periodic position.		
3	**5.Carbohydrate Chemistry:** Sources of carbohydrates and photosynthesis. Detailed classification of carbohydrates: classification, isomerism and reactions of monosaccharides. A brief introduction of disaccharides (e.g., Sucrose, maltose, lactose) and Polysaccharides (e.g., starch, and cellulose). Economic and biological importance of carbohydrates.	Lecture, Group Discussion, Assignment	2, 3, 4, 5
3	**6.Chemistry of Amino acid, proteins Enzyme:** Definition, Source, Structures, classification, synthesis physical and chemical properties of amino acids. Brief study on peptide and proteins. Biological and industrial importance, application of amino acids, proteins & enzymes. Co-enzyme: Classification, structure and mechanism and their role.	Lecture, Group Discussion, Assignment	2, 3, 4
2	**7. Hormones:** classification, general characteristics of hormone action. Functions of different hormone - thyroid, pituitary, pancreatic and hormones of the adrenal glands, their physiological roles and disease states.	Lecture, Group Discussion, Assignment	1, 3
3	**8.Polymer chemistry:** Polymers and macro molecules, Detailed classification of polymers, basic structure of polymers, Copolymers; alternating, random, block and graft copolymers. Industrial polymers: rubber, elastomer, plastics and fibers. Natural, synthetic and biopolymers, Application of polymers and biopolymers.	Lecture, Group Discussion, Assignment	4, 5
4	**9.Pollution and Pollutants:** Pollution, environmental pollution environmental pollution (air, water & soil pollution), pollutant, classification of pollutants, types of pollution PCBS and their sources and hazards, Detection & estimation of PCBS. Bio-multifaction. measurements of important parameters such as PH, DO, BOD, COD and temperature for water quality assessments. effects of industrial effluents, effects of urban wastes, control of soil pollution.	Lecture, Group Discussion, Assignment	1, 6
3	**10.Heavy metals in the Environment:** trace metals, light metals and heavy metals, deadly heavy metals, sources of heavy metals, biochemical effects, toxicity, toxicology, control and treatment of mercury, chromium, arsenic and lead.	Lecture, Group Discussion, Assignment	1, 2

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Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Bahl, B. S., & Bahl, A. (2024). *Text- Book of Organic Chemistry*. S. Chand & Company Ltd Ram Nagar, New Delhi-1I005S

Darrell, E. and Gammon, S. D. (11th Edition). *General Chemistry*, Published by Mc Grew Hill.

De, A. K. (2003). *Environmental Chemistry.* Publisher, New Age International. ISBN, 8122414885, 9788122414882

English, J., & H.G Cassidy and Baird, R. I. (1949). *Principles of Organic Chemistry: An Introductory Text in Organic Chemistry*. MeGraw Hill.

Finar, I. L. (1973). *Organic Chemistry, Vol. I*. Pearson Education India.

Griffin, R. W. (2024).*Modern Organic Chemistry*. McGraw Hill.

Haider, S. Z. (2024 Reprint). *Selected Topics on Advanced Inorganic Chemistry*, Students' publication.

Morrison, R.T. and Boyd, R. N. (1992). *Organic Chemistry*. Benjamin-Cummings Pub Co

Muny, J. Mc. (2010). *Organic Chemistr.* Thomson Broks-Coole.

Benjamin, Inc., Menlo Park, CA. ISBN 0-8053-8329-8.

Solomons, T. W. G. (2008). *Fundamental of Organic Chemistry*. Wiley, ISBN: 0471146498, 9780471146490

PROVISIONAL

Course Code: 8222861	**Marks: 50**	**Credits: 2**	**Class Hours: 28**
Course Title: Environmental Chemistry			

Course Description

This course provides a comprehensive introduction to the chemical principles governing the natural environment and the impact of anthropogenic activities on earth systems. It encompasses a detailed study of the structural segments of the environment—atmosphere, hydrosphere, lithosphere, and biosphere—and focuses on the chemical pathways, transport dynamics, and transformations of pollutants. Students will explore the mechanisms behind critical global phenomena, including acid rain, the greenhouse effect, ozone depletion, and bio-magnification. The curriculum emphasizes quantitative water quality metrics (pH, DO, BOD, COD), modern wastewater treatment processes, soil degradation vectors, and the biochemical toxicology of toxic persistent compounds like PCBs and heavy metals. Through theoretical frameworks, case studies, and numerical assessments, this course equips students with the essential analytical tools needed to evaluate environmental degradation and formulate sustainable mitigation strategies.

Course Objectives

Upon successful completion of this course, students will be able to:

- **Develop** a fundamental understanding of the chemical composition, interactions, and structural layers defining environmental segments.
- **Analyze** the chemical reactions, source identification methods, and transport pathways of hazardous air, water, and soil pollutants.
- **Evaluate** industrial and municipal water contamination by measuring key water quality parameters and understanding standard purification technologies.
- **Explain** the biochemical mechanisms of toxicity, bioaccumulation, and environmental hazards associated with heavy metals and persistent organic compounds.
- **Apply** physical and chemical principles to design sustainable control mechanisms for industrial effluents, urban waste management, and soil remediation.
- **Cultivate** scientific reasoning, problem-solving abilities, and an ethical foundation required to address domestic and global environmental challenges.

Course Learning Outcomes (CLOs)

By the end of this course, students will be able to:

CLO	Course Learning Outcome	Bloom's Level
CLO 1	**Define** and **explain** the fundamental segments of the environment and their systemic interactions	*Understand*
CLO 2	**Analyze** the sources, chemical behavior, and global consequences of air, water, and soil pollutants.	*Analyze*

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CLO 3	**Evaluate** wastewater and municipal water treatment parameters (pH, DO, BOD, COD).	*Evaluate*
CLO 4	**Assess** the biochemical toxicity and control methods for heavy metals and hazardous compounds like PCBs.	Evaluate

Course Contents

Week	Topics	Teaching-Learning Strategy	Assessment Strategy	CLO
1-3	**1. Environment:** Introduction components of environment, factors affecting environment, environmental management, environment and health, environmental chemistry, segments of environment – atmosphere, hydrosphere, lithosphere and biosphere, structure of atmosphere.	Interactive Lecture, Group Discussion, Concept Mapping	Quiz, Class Test, Short Assignment	**CLO1**
4-7	**2. Pollution and Pollutants:** Pollution, environmental pollution, pollutant, classification of pollutants, types of pollution. PCBs and their sources and hazards, Detection & estimation of PCBs. Bio-magnification/Biomultification.	Lecture, Case Study Analysis, Group Discussion	Class Test, Assignment, Viva-voce	**CLO2 CLO4**
8-14	**3. Air Pollution:** Introduction, air quality, major sources of air pollution, gaseous pollutants. Acid rain-how acid rain is formed, adverse effects of acid rain. Greenhouse effect- how the greenhouse effect is produced, consequences of greenhouse effect and global warming. El Niño phenomenon and its effect. Ozone depletion, mechanism of ozone depletion, effects of ozone depletion.	Lecture, Multimedia Presentation, Problem-Based Learning	Quiz, Mid Term Exam, Assignment	**CLO2**
15-19	**4. Water Pollution:** Introduction, classification of water pollutants. Physical, chemical and biological characteristics of wastewater. Industrial wastewater treatment, municipal water treatment. Water quality parameters and standards, measurements of important parameters such as pH, DO, BOD, COD and temperature for water quality	Interactive Lecture, Numerical Exercises, Laboratory Demonstration	Class Test, BOD/COD Problems, Report Writing	**CLO3**

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Week	Topics	Teaching-Learning Strategy	Assessment Strategy	CLO
	assessments.			
20-24	**5. Soil Pollution:** Composition of soil, importance of soil to the biosphere, sources of soil pollution. Effects of soil pollution-synthetic fertilizer and pesticides, effects of industrial effluents, effects of urban wastes, control of soil pollution.	Lecture, Case Study Analysis, Group Discussion	Quiz, Assignment	**CLO2**
25-28	**6. Heavy metals in the Environment:** Trace metals, light metals and heavy metals, deadly heavy metals, sources of heavy metals, biochemical effects, toxicity, toxicology, control and treatment of mercury, chromium, arsenic and lead.	Lecture, Problem-Based Learning, Course Revision	Final Assessment, Viva-voce	**CLO4**

Course Assessment Framework:

Total Course Marks: 50 Exam Duration: 2.5 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	4%
	In-course Examination	10%
	Quiz / Assignment Presentation	6%
Summative Assessment (80%)	Final Examination • Students will have to answer 4 questions out of 6 • 10 marks per question ($4 \times 10 = 40$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Dara, S. S. (2010). *A textbook of environmental chemistry and pollution control*. S. Chand & Company Ltd.

De, A. K. (2010). *Environmental chemistry* (7th ed.). New Age International Publishers.

Manahan, S. E. (2022). *Environmental chemistry* (11th ed.). CRC Press.

Sharma, B. K. (2014). *Environmental chemistry*. Goel Publishing House.

PROVISIONAL

Course Code	8223057	Marks:100	Credits: 4	Class Hours: 56
Course Title	Botany - II			

Course Description

This course provides an integrated understanding of fundamental concepts in cell biology, genetics, plant breeding, biotechnology, ecology, conservation biology, plant physiology, and phytochemistry. It examines cellular organization and division, inheritance and genetic variation, plant improvement and evolution, biotechnological applications, ecosystem structure and function, biodiversity conservation, plant physiological processes, and major plant biomolecules. The course emphasizes the application of botanical knowledge to plant growth, crop improvement, environmental management, biodiversity conservation, and sustainable development.

Course Objectives (COs)

After successful completion of this course, students will be able to:

1. Develop a comprehensive understanding of cell biology, genetics, plant breeding, and evolutionary principles underlying plant growth, reproduction, inheritance, and variation.
2. Explain the fundamental principles and applications of biotechnology in agriculture, industry, environmental management, plant improvement, and sustainable resource utilization.
3. Develop an understanding of ecological processes, environmental issues, ecosystem functioning, and biodiversity conservation, with emphasis on sustainable management of plant resources.
4. Apply knowledge of plant physiology and phytochemistry to explain plant growth, metabolism, productivity, adaptation, and the importance of major plant biomolecules

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome (CLO)	Bloom's Taxonomy Level
CLO1	Explain the ultrastructure of plant cells, chromosome organization, and the mechanisms and significance of mitosis and meiosis.	Understand

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CLO2	Explain the fundamental principles of inheritance, linkage, crossing over, mutation, and their roles in plant genetics and crop improvement.	Understand
CLO3	Describe the principles and applications of plant breeding, artificial pollination, selection methods, and theories of evolution.	Understand
CLO4	Explain the basic concepts and applications of biotechnology, including microbial fermentation, dairy biotechnology, and biogas production.	Understand
CLO5	Analyze ecological principles, ecosystem structure and function, ecological adaptations, biogeochemical cycles, environmental pollution, and greenhouse effects.	Analyze
CLO6	Explain the importance of biodiversity conservation and evaluate in-situ and ex-situ conservation strategies for sustainable management of plant resources.	Evaluate
CLO7	Explain the physiological processes of plants, including water relations, mineral nutrition, photosynthesis, respiration, enzyme action, photoperiodism, vernalization, and plant hormones.	Understand
CLO8	Describe the classification, structure, biological functions, and economic importance of major plant carbohydrates and communicate scientific concepts effectively through assignments and examinations.	Apply

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Mapping of CLOs with PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2			√	√								
CLO3					√	√	√					
CLO4								√	√	√		
CLO5										√	√	√
CLO6	√	√	√									
CLO7				√	√							
CLO8						√	√					

Course Content

Unit Title	Detailed Unit Contents	Teaching-Learning Strategies	Assessment	CLO
Unit 1. Cytology **Weeks 1–4**	Ultra-structure of a eukaryotic cell, differences between prokaryotic and eukaryotic cells, physical and chemical composition of cell wall and plasma membrane. **Chromosome:** Physical and chemical nature, chromosome organization, types of chromosomes and role in heredity. **Cell division:** Mitosis and meiosis, stages and significance in growth, reproduction, genetic variation and evolution. **Cell organelles:** Structure and functions of nucleus, mitochondria, chloroplast, endoplasmic reticulum, ribosome, Golgi apparatus, vacuole and other cellular components.	Interactive lectures, multimedia presentations, classroom discussion, diagrams	Quiz, class test, assignment, final examination	CLO1
Unit 2. Genetic **Weeks 5–8**	**Basic concepts of genetics:** Gene, DNA and RNA structure, DNA replication, gene expression, Mendel's laws; linkage and crossing over; mutation, point mutation, physical and chemical mutagens and significance.	Lectures, multimedia presentations	Quiz, assignment, midterm, final examination	CLO1, CLO2
Unit 3. Plant Breeding and Evolution **Weeks 9–11**	**Plant Breeding and Evolution:** Definition and significance of plant breeding. Method of artificial pollination in plants. Selection method.	Lectures, case studies, group discussion	Class presentation , assignment, written examination	CLO2

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	Theories of evolution: Darwin's theory and synthetic theory of evolution.			
Unit 4. Biotechnology **Weeks** **12–15**	Definition, concept, history and scope of biotechnology; applications in agriculture, industry, medicine and environmental management. **Milk products:** Techniques of production of curd and butter; role of microorganisms in dairy fermentation. **Biogas technology:** Production methods, microbial role, composition of biogas and its uses as renewable energy. **Plant tissue culture and its applications:** Principles, techniques, micropropagation and importance in crop improvement and conservation. **Biofertilizers and biopesticides:** Types, production, applications and importance in sustainable agriculture.	Interactive lectures, demonstrations, case studies	Quiz, assignment, final examination	CLO2
Unit 5. Plant Ecology and Environmental Science **Weeks 16–20**	Definition, differences between autecology and synecology. **Ecological factors:** Climatic, edaphic and biotic. **Ecological adaptation:** Characteristics of hydrophytes, xerophytes and halophytes. **Ecosystem:** Definition; pond ecosystem; food chain; food web; ecological pyramids.	Lectures, field examples, case studies, group discussion	Assignment, presentation, midterm, final examination	CLO3

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	Biogeochemical cycles: Nitrogen and carbon cycles. **Pollution:** Air and water pollution. **Greenhouse effects:** Definition, sources of greenhouse gases; causes and effects of greenhouse gases; CO_2, CFC, CH_4 and N_2O; ozone layer depletion; carbon dioxide and the world climate; control of greenhouse effects.			
Unit 6. Conservation Biology	**Conservation:** Importance of conservation of plant species. Methods of ex-situ and in-situ conservation.	Interactive lecture, case studies	Assignment, written examination	CLO3
Unit 7. Plant Physiology and Phytochemistry **Weeks 21–28**	**Water relation:** Mechanisms of absorption of water and minerals. **Mineral nutrition.** **Photosynthesis:** General account and modern concept. **Respiration:** Aerobic and anaerobic. **Enzymes:** Definition and properties. **Photoperiodism.** **Vernalization:** Definition and mechanism. **Plant hormones:** A general account. **Carbohydrates:** Classification with examples; structures of important monosaccharides and oligosaccharides and isomerism; storage polysaccharides; starch, inulin, glycogen; structural polysaccharides; cellulose and economic importance of carbohydrates.	Lectures, tutorials, problem-solving, multimedia presentations	Quiz, class test, assignment, midterm, final examination	CLO4, CLO5

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Course Assessment Framework:

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Agrios, G. N. (1997). *Plant Pathology* (4th ed.). Academic Press, London, UK.

Lewin, B. (2000). *Gene 2000*. Oxford University Press & Cell Press, Oxford, UK.

Devlin, M. R., & Witham, H. F. (1986). *Plant Physiology* (4th ed.). CBS Publishers and Distributors, New Delhi, India.

Esau, K. (1953). *Plant Anatomy*. John Wiley & Sons, Inc., New York, USA.

Kumar, H. D. (1995). *General Ecology*. Vikas Publishing House, New Delhi, India.

Pelczar, M. J., Chan, E. C. S., & Krieg, N. R. (1993). *Microbiology: Concepts and Applications.* McGraw-Hill Book Company, New York, USA.

Vashishta, P. C. (1993). *Botany for Degree Students: Pteridophyta*. S. Chand & Company Ltd., New Delhi, India.

আখতারুজ্জামান, ম. (২০০০). *বিবর্তনবাদ*, হাসান বুক হাউজ, ঢাকা।

কর্মকার, যদুলাল. (২০০০) *উদ্ভিদ শরীর বিজ্ঞান*, হাসান বুক হাউজ, ঢাকা।

খন্দকার এম. (১৯৯৪). *ডলমনোলজী*, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা, ঢাকা।

বাসার, এম. এ., এম.এ. হাসান এবং ম. রফিকুল ইসলাম. (২০০৪). *উদ্ভিদ বিজ্ঞান*, হাসান বুক হাউজ, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. (১৯৯৬). *উংলাদেশের ভেষজ উদ্ভিদ*, আশরাফিয়া বই ঘর, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. এবং এম. কে. আলম. (১৯৯৭). *উদ্ভিদ শ্রেণী বিন্যাস তত্ত* (৩য় সংস্করণ), হাসান বুক হাউস, ঢাকা।

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Course Code	8223058	Marks:50	Credits: 2	Class Hours: 42
Course Title	Botany -II Practical			Exam Time: 6 hours

Course Description

This practical course develops fundamental skills in the identification, classification, and study of major plant groups, including algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms. It provides hands-on experience in specimen collection, dissection, mounting, microscopy, scientific drawing, plant physiology experiments, mitotic cell division, and ecological adaptation studies. The course emphasizes accurate observation, laboratory techniques, scientific documentation, specimen identification, and interpretation of botanical characteristics and their ecological significance.

Course Objectives (COs)

After completion of this course, students will be able to:

1. Develop practical skills in the identification, classification, and study of representative algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms through dissection, mounting, observation, and illustration.
2. Explain the morphological and anatomical characteristics of different plant groups and relate their structural adaptations to their ecological habitats.
3. Perform basic physiological experiments to understand fundamental plant processes such as plasmolysis, osmosis, and photosynthetic oxygen evolution.
4. Observe and identify different stages of mitotic cell division using plant materials and understand their significance in growth and development.
5. Develop skills in scientific observation, laboratory techniques, record keeping, scientific drawing, and interpretation of botanical specimens.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome	Bloom's Taxonomy Level
CLO1	Identify and classify representative genera of algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms through practical study.	Understand
CLO2	Perform dissection, mounting, microscopic observation, scientific drawing, and description of plant specimens.	Apply

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CLO3	Conduct basic plant physiology experiments and interpret the results related to water relations and photosynthesis.	Analyze
CLO4	Identify different stages of mitosis and explain their importance in plant growth and development.	Understand
CLO5	Analyze ecological adaptations of hydrophytic and xerophytic plants through morphological and anatomical observations.	Analyze
CLO6	Maintain laboratory records, prepare scientific reports, and demonstrate botanical knowledge through viva-voce.	Apply

Mapping of CLOs with PLOs

CLOs	PLO 1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√	√								
CLO3				√	√	√	√					
CLO4	√	√		√				√	√			
CLO5		√				√	√		√	√		
CLO6	√	√	√								√	
CLO7				√	√					√		
CLO8					√	√						

Course Content

Unit Title	Detailed Unit Contents	Teaching–Learning Strategies	Assessment	CLO(s)
Unit 1: Plant Diversity, Identification and Systematics	1. Detailed study including dissection (where necessary), mounting, drawing, description and identification with classification of the following genera: Algae: *Chlamydomonas*, *Oedogonium* and *Vaucheria*. Fungi: *Yeast* and *Ascobolus*. Bryophyte: *Riccia* and *Marchantia*.	Demonstration of specimens, laboratory observation, microscopic study, dissection and temporary slide preparation, scientific drawing practice, guided identification exercises.	Practical examination, specimen identification, laboratory performance, scientific drawing, practical record assessment.	CLO1 CLO2

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	Pteridophyte: S*elaginella*. Gymnosperms: *Cycas*. Angiosperm: Poaceae and Fabaceae.			
Unit 2: Identification of Plant Genera with Diagnostic Characters	2. Identification of the following genera with reasons: Algae: *Volvox*, *Polysiphonea* and *Fucas*. Fungi: *Agaricus*, *Puccinia* and *Penicillium*. Lichen: *Crustose*, *Foliose* and *Fructose*. Bryophyte: *Anthoceros*. Pteridophyte: *Psilotum*, *Azolla* and *Pteris*. Gymnosperms: Male and female cones of *Cycas*. Angiosperms: Scientific names of common plants.	Specimen display, field-based observation, herbarium study, comparative analysis, identification keys and discussion.	Spot identification test, viva-voce, practical examination, herbarium/specimen report evaluation.	CLO1 CLO2
Unit 3: Plant Physiology Experiments	Experimental study of basic physiological processes: **(a)** Demonstration of plasmolysis: principle, experimental procedure, observation and precautions. **(b)** Demonstration of osmosis: experimental setup, observation and interpretation. **(c)** Evolution of oxygen during photosynthesis: experimental demonstration and factors affecting photosynthetic activity.	Laboratory demonstration, hands-on experimentation, observation recording, data interpretation and discussion of physiological processes.	Experimental performance, observation note, practical report, viva-voce.	CLO3
Unit 4: Cytology and Cell Division	Study of different stages of mitosis in onion (*Allium cepa*) root tip: Preparation of root tip squash. Identification	Microscopic observation, slide preparation demonstration, image-	Microscopic identification, slide preparation	CLO4

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	of stages: prophase, metaphase, anaphase and telophase. Significance of mitosis in plant growth and development.	based learning, identification and interpretation of mitotic stages.	assessment, practical examination and viva.	
Unit 5: Plant Ecology and Adaptation Studies	**(a)** Morphological and anatomical studies of hydrophytic and xerophytic plants. **(b)** Ecological identification of plants: Identification of plants according to habitat. Scientific naming and ecological significance.	Field observation, specimen collection, ecological comparison, anatomical study, group discussion and field-based learning.	Field report, plant identification test, practical record evaluation, viva-voce.	CLO5
Unit 6: Laboratory Documentation and Scientific Communication	Preparation and maintenance of laboratory notebook including: Recording observations. Scientific drawings and labeling. Classification and description of specimens. Preparation of practical reports.	Guided note preparation, feedback on scientific documentation, demonstration of proper laboratory practices.	Laboratory notebook evaluation, report assessment, continuous assessment.	CLO6
Unit 7: Viva-Voce and Practical Competency Evaluation	**Viva-Voce:** Assessment of students' understanding of: Plant identification and classification. Laboratory techniques. Experimental principles and observations. Ecological and cytological concepts.	Interactive questioning, specimen-based discussion, problem-solving and conceptual assessment.	Final practical examination, viva-voce, overall laboratory performance assessment.	CLO6

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List of Books

Agrios, G.N. (1997). *Plant Pathology* (4th ed.). Academic Press, London.

Benjamin L. (2000). *Gene 2000*. Oxford University Press and Cell Press. Bold, H.C. and

Kumar, H.D. (1995). *General Ecology*, Vikash Pub. House, India.

Lawrence, G.H.M. (1951). *Taxonomy of Vascular Plants*. The Macmillan Co. New York.

Pelczer, M. J., Chan, E. C. S. & Krieg, N. R. (1993). *Microbiology: Concepts and Applications*. McGraw Hill Book Co. Inc. New York.

Vashista, P.C. 1993). *Botany for Degree Students: Pteridophyta.* S.C. Chand & Co. Ltd. Ramnagar, New Delhi.

আখতারুজ্জামান, ম. (২০০০). *বিবর্তন বাদ*, হাসান বুক হাউজ, ঢাকা।

কর্মকার, যদুলাল. (২০০০). *উদ্ভিদ শরীর বিজ্ঞান*, হাসান বুক হাউজ, ঢাকা।

খন্দকার মনিরুজ্জামান, (১৯৯৪). *লিমনোলজী*, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা, ঢাকা।

বাসার, এম. এ., এম.এ. হাসান এবং ম. রফিকুল ইসলাম. (২০০৪). *উদ্ভিদ বিজ্ঞান*, হাসান বুক হাউজ, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. (১৯৯৬). *বাংলাদেশের ভেষজ উদ্ভিদ*, আশরাফিয়া বই ঘর, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. এবং এম. কে. আলম. (১৯৯৭). *উদ্ভিদ শ্রেণী বিন্যাস* তত্ত (৩য় সংস্করণ), হাসান বুক হাউস, ঢাকা।

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Instruction to the Examiners

Time: 6 hours Marks: 50

1. Detailed Study, Dissection, Mounting, Description and Identification of Plant Genera

Marks: 12 (Slide preparation/mounting: 2+ Neat labeled diagrams: 2+ Identification: 0.5+ Classification and identifying characters: 1.5=6×2=12)

Specimen A and B:
Two specimens will be provided from the following groups:

- Algae: *Chlamydomonas, Oedogonium, Vaucheria*
- Fungi: Yeast, *Ascobolus*
- Bryophyta: *Riccia, Marchantia*
- Pteridophyta: *Selaginella*
- Gymnosperms: *Cycas*
- Angiosperms: Families Poaceae and Fabaceae

Students will perform:

- Dissection (where necessary)
- Preparation of slides/mounting
- Observation and identification
- Classification
- Description of diagnostic features
- Scientific drawings with proper labeling

2. Spot Identification of Plant Genera with Reasons Marks: 9

(Identification: 0.5+ Identifying Characteristics: 1.0=1.5×6=9)

Specimens C–H will be provided for identification from the following groups:

Algae: *Volvox, Polysiphonia, Fucus*; Fungi: *Agaricus, Puccinia, Penicillium*

Lichens: Crustose, Foliose, Fruticose; Bryophyta: *Anthoceros*

Pteridophyta: *Psilotum, Azolla, Pteris*

Gymnosperms: Male and female cones of *Cycas*;

Angiosperms: Common flowering plants with scientific names

3. Plant Physiology Experiments Marks: 6

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(Experimental setup and procedure: 3+ Observation and interpretation: 2 + Precautions and conclusion: 1= 6)

Students will perform experiments related to:

(a) Demonstration of Plasmolysis: Principle, Experimental procedure, Observation, Precautions

(b) Demonstration of Osmosis: Experimental setup, Observation and interpretation

(c) Evolution of Oxygen During Photosynthesis: Experimental demonstration, Observation of oxygen release, Factors affecting photosynthetic activity

4. Cytology: Study of Mitosis in Onion Root Tip Marks: 6

(Preparation of slide: 2+ Identification of mitotic stages: 2+ Neat labelled diagrams: 1+ Significance/comment:1=06)

Students will prepare and study onion (*Allium cepa*) root tip squash.

Stages to be studied: Prophase, Metaphase, Anaphase, Telophase

5. Plant Ecology and Adaptation Study Marks: 4

Specimen I and J (Identification:0.5+ Identifying characters/adaptive features:1.5=2×2=4)

Specimens will be selected from:

- Hydrophytic plants
- Xerophytic plants

Students will study:

- Morphological adaptations, Anatomical adaptations
- Ecological significance, Identification of plants with scientific names

6. Collection, Laboratory Record and Viva-Voce	**Marks: 13**
Components	Marks
Collection of plant specimens (Algae, Fungi, Bryophyta, Pteridophyta, Gymnosperms and Angiosperms)	3.0
Practical Laboratory Notebook	5.0
Viva-Voce	5.0
Total	13.0

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Total Marks Distribution

Examination Component	Marks
Detailed study of specimens (A & B)	12
Spot identification (C–H)	9
Plant physiology experiment	6
Cytology (Mitosis)	6
Ecology and adaptation study	4
Collection	3
Practical Notebook	5
Viva-Voce	5
Grand Total	50

General Instructions

- Students must submit properly prepared practical notebooks containing classification, descriptions, observations and labelled drawings.
- Slides must be prepared carefully following standard laboratory techniques.
- Identification answers must include appropriate diagnostic characters.
- Scientific names should be written according to accepted botanical nomenclature.
- Herbarium specimens and field collections must include proper labels.
- Viva-voce will evaluate students' understanding of practical techniques, specimen identification and botanical concepts.

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Course Code	8223358	Marks:50	Credits: 2
Course Title	**Viva-Voce (Soil Science)**		

1. Course Description

The Viva-Voce course is designed to comprehensively assess students' conceptual depth, analytical capabilities, communication clarity, and practical competencies acquired during their second year. The oral examination directly evaluates the students' synthesis of **Soil Physics-I, General Microbiology, Soil Mineralogy and Colloids**, and their corresponding **Practical Laboratory Applications**.

Furthermore, the course gauges students' ability to link fundamental soil science concepts with contemporary global issues such as food security, environmental sustainability, climate change resilience, and the United Nations Sustainable Development Goals (SDGs 4, 6, 13, and 15) within the context of Bangladesh.

2. Course Objectives

Core Assessment: Evaluate the depth of understanding of second-year core Soil Science courses.
Practical Application: Verify the students' technical grasp of field sampling, lab instrumentation, and data interpretation.
Critical Thinking: Measure the ability to diagnose land degradation, soil degradation, and nutrient deficiencies using scientific reasoning.
Communication & Professionalism: Assess scientific terminology usage, oral presentation flow, confidence, and academic ethics.
Global Alignment: Examine the students' awareness of soil resources as critical assets for national food security and climate resilience.

3. Course Learning Outcomes (CLOs)

Upon successful completion of this viva-voce course, students will be able to:

CLO 1: Articulate primary soil physical properties, water regimes, thermal attributes, and gaseous exchanges clearly using professional terminology.
CLO 2: Categorize microbial diversity, metabolic pathways, and describe the biochemical dynamics of the soil biomass habitat.
CLO 3: Explain the crystalline structure, charge characteristics, weathering sequences, and adsorption behaviors of primary/secondary soil minerals and colloids.
CLO 4: Demonstrate an understanding of the working principles, calculations, and error management of key laboratory procedures (e.g., pH, CEC, Nitrogen distillation, mechanical analysis, and bacterial staining).
CLO 5: Connect soil science interventions to real-world national problems like soil salinity, erosion, and agricultural adaptation under climate change pressures.

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4. Viva-Voce Preparation Modules & Core Question Bank

The viva preparation will focus on four academic pillars based on the core curriculum:

Pillar 1: Soil Physics – I

- **Key Topics:** Particle size distribution, texturing methods, bulk vs. particle density, soil structure genesis, Atterberg's constants, gas transport mechanisms, thermal conductivity, and soil water potentials.
- **Sample Viva Questions:**
  - How does high bulk density affect root penetration and soil aeration?
  - Can you explain the practical agricultural significance of the *Plastic Limit* and *Liquid Limit* (Atterberg's constants)?
  - Differentiate between gravitational water, capillary water, and hygroscopic water in terms of soil water potential.
  - What mechanisms govern gaseous exchange between the atmosphere and the soil matrix?

Pillar 2: General Microbiology

- **Key Topics:** Whittaker's five-kingdom concept, soil as a bioenergy hub, microscopic physics, pure culture isolation, microbial control, carbohydrate metabolism, and enzyme kinetics
- **Sample Viva Questions:**
  - Why is the soil rhizosphere considered the primary hub of bioenergy for soil biota?
  - Explain the significance of the Michaelis-Menten constant regarding soil enzyme activities.
  - What is the difference between physical sterilization and chemical disinfection in microbial control?
  - Describe the pathway variations between oxidative and fermentative microbial metabolism.

Pillar 3: Soil Mineralogy and Colloids

- **Key Topics:** Silicate structures weathering sequences, permanent vs. variable charge, Helmholtz/Gouy-Chapman double layer models, zeta potential, Donnan membrane equilibrium, and adsorption isotherms (Langmuir & Freundlich).
- **Sample Viva Questions:**
  - Explain how isomorphous substitution leads to the development of permanent negative charges in 2:1 clay minerals.
  - Compare the structural configurations and swelling behaviors of Kaolinite versus Smectite.
  - What is the "Zeta Potential," and how does it influence soil colloidal flocculation and dispersion?
  - How do Langmuir and Freundlich adsorption isotherms help us understand nutrient retention in soils?

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Pillar 4: Soil Science Practical & Instrumentation

- **Key Topics:** Glass electrode pH meters, Kjeldahl digestion/distillation, mechanical analysis (hydrometer/pipette methods), ammonium acetate extraction for CEC, and gram staining.
- **Sample Viva Questions:**
  - Walk me through the chemical principles behind determining Cation Exchange Capacity (CEC) using ammonium acetate.
  - Why do we use a dispersing agent (like sodium hexa-metaphosphate) during soil particle-size mechanical analysis?
  - What are the main steps and active reagents used in determining Total Nitrogen using the Kjeldahl method?
  - What color do Gram-negative bacteria retain during staining, and what structural feature causes this?

5. Teaching & Preparation System

To prepare students effectively, the department will organize structured pre-viva sessions:

- **Interactive Reviews:** Peer-led question-and-answer circles targeting core definitions.
- **Mock Viva:** Simulated examination boards featuring immediate feedback on posture, vocal clarity, technical accuracy, and stress management.
- **Lab Walkthroughs:** On-site laboratory reviews of instrumentation, chemical handling safety, and calculation formulations.

6. Assessment Framework & Evaluation Criteria

The final 50-mark score will be collectively determined by an external and internal Examination Board using the following rubric components:

Evaluation Component	Description
Conceptual Clarity	Accuracy of scientific definitions, theories, and concepts across all modules.
Practical Mastery	Ability to explain laboratory procedures, calculations, and data interpretation.
Critical Application	Linking soil principles to environmental issues and climate scenarios in Bangladesh.
Communication Skills	Use of appropriate terminology, structure of answers, confidence, and professional demeanor.

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